

Systems Biology: Mathematics for Biologists



Levien van Zon, Theoretical Biology, UU

Efficient analysis of systems of two differential equations

Summary of previous chapters

Given a general 2D system:

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y) \end{cases}$$

with an equilibrium at (x^*, y^*) .

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The eigenvalues $\lambda_{1,2}$ give the equilibrium type and stability.

Trace and Determinant

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$\det A = ad - bc$ is the **determinant** of a matrix.

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into:

$$\lambda^2 - \operatorname{tr} A \lambda + \det A = 0$$

Rewriting the eigenvalues

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can be written as:

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with solutions:

$$\lambda_{1,2} = \frac{\operatorname{tr} A \pm \sqrt{D}}{2}$$

whereby $D = (\operatorname{tr} A)^2 - 4 \times \det A$

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Also note that $\lambda_1 + \lambda_2 = \operatorname{tr} A$, and $\lambda_1 \times \lambda_2 = \det A$.

Determinant Trace Method

Finding the equilibrium type without finding the eigenvalues:
the **determinant trace method**.

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 - if $\operatorname{tr} A < 0 \rightarrow$ stable node
- $\det A > 0$ and $D < 0$. We have complex eigenvalues with $\operatorname{tr} A$ as real part.

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 - if $\operatorname{tr} A > 0 \rightarrow$ unstable spiral

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 - if $\operatorname{tr} A > 0 \rightarrow$ unstable spiral
 - if $\operatorname{tr} A < 0 \rightarrow$ stable spiral

Determinant Trace Method

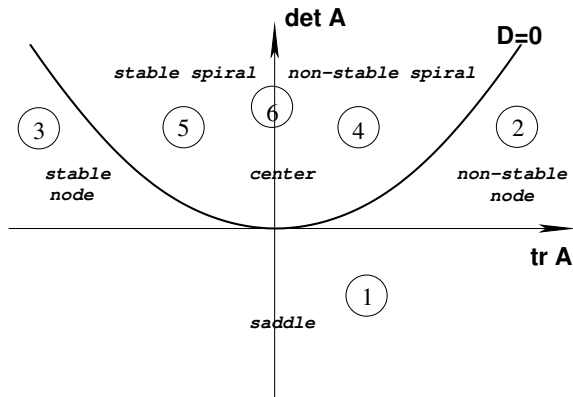
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- $\det A > 0$ and $D < 0$. We have complex eigenvalues with $\operatorname{tr} A$ as real part.
 - if $\operatorname{tr} A > 0 \rightarrow$ unstable spiral
 - if $\operatorname{tr} A < 0 \rightarrow$ stable spiral
 - if $\operatorname{tr} A = 0 \rightarrow$ center point

Determinant Trace Method

Graphical representation:



Example 1

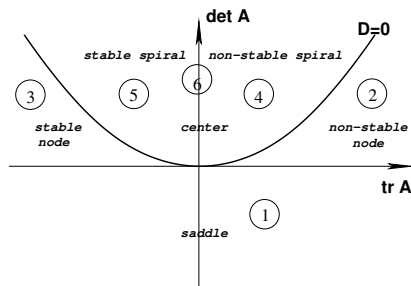
$$\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Example 2

$$\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} -2 & 3 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Graphical Jacobian

Consider again:



It is sufficient to know **the sign of $\text{tr } A$ and $\text{det } A$** to find whether an equilibrium is a saddle, a stable spiral/node or an unstable spiral/node.

Graphical Jacobian

Use the vectorfield to determine the sign of the Jacobian elements:

$$J = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}$$

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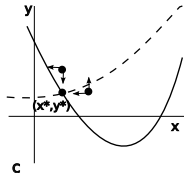
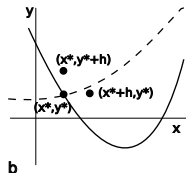
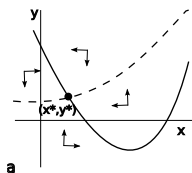
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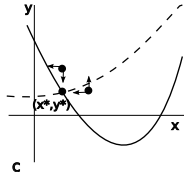
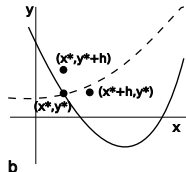
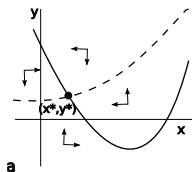
Graphical Jacobian

Example



Graphical Jacobian

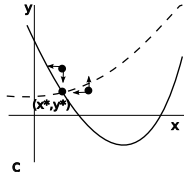
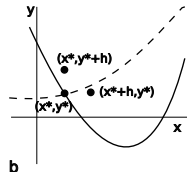
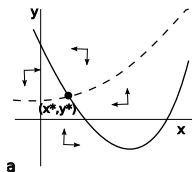
Example



$$\frac{\partial f}{\partial x} = -$$

Graphical Jacobian

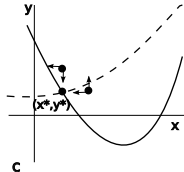
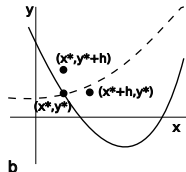
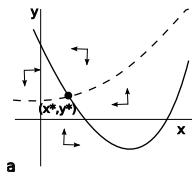
Example



$$\frac{\partial f}{\partial x} = -$$
$$\frac{\partial f}{\partial y} = -$$

Graphical Jacobian

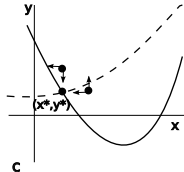
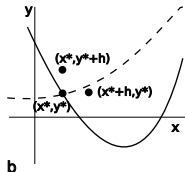
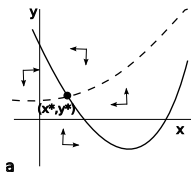
Example



$$\begin{aligned}\frac{\partial f}{\partial x} &= - \\ \frac{\partial f}{\partial y} &= - \\ \frac{\partial g}{\partial x} &= +\end{aligned}$$

Graphical Jacobian

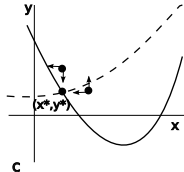
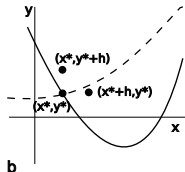
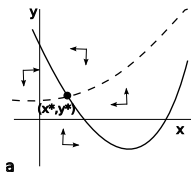
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Graphical Jacobian

Example

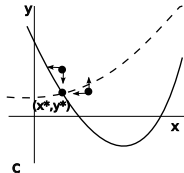
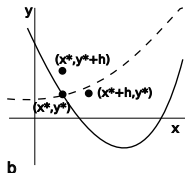
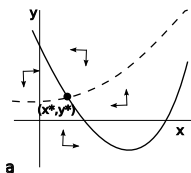


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$$J = \begin{pmatrix} -\alpha & -\beta \\ +\gamma & -\delta \end{pmatrix}$$

Graphical Jacobian

Example



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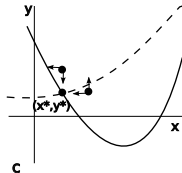
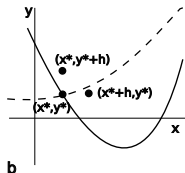
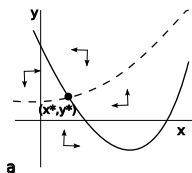
Therefore

$$\det J = (-\alpha \times -\delta) - (-\beta \times \gamma)$$

$$J = \begin{pmatrix} -\alpha & -\beta \\ +\gamma & -\delta \end{pmatrix}$$

Graphical Jacobian

Example



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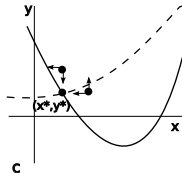
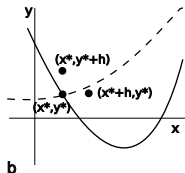
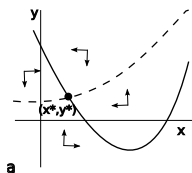
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$$\det J = (-\alpha \times -\delta) - (-\beta \times \gamma) = \alpha\delta + \beta\gamma$$

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Graphical Jacobian

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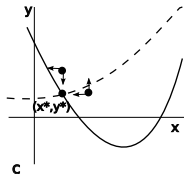
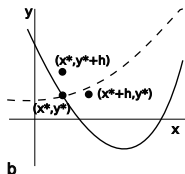
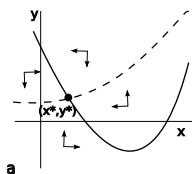
Therefore

$$\det J = (-\alpha \times -\delta) - (-\beta \times \gamma) = \alpha\delta + \beta\gamma > 0$$

$$J = \begin{pmatrix} -\alpha & -\beta \\ +\gamma & -\delta \end{pmatrix}$$

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Example



$$\begin{aligned}\frac{\partial f}{\partial x} &= - \\ \frac{\partial f}{\partial y} &= - \\ \frac{\partial g}{\partial x} &= + \\ \frac{\partial g}{\partial y} &= -\end{aligned}$$

Therefore

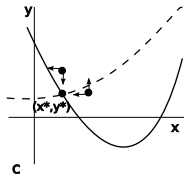
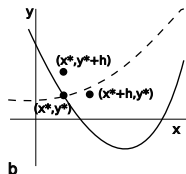
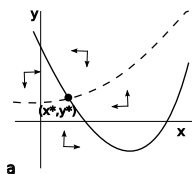
$$\det J = (-\alpha \times -\delta) - (-\beta \times \gamma) = \alpha\delta + \beta\gamma > 0$$

$$J = \begin{pmatrix} -\alpha & -\beta \\ +\gamma & -\delta \end{pmatrix} \quad \text{and}$$

$$\text{tr } J = -\alpha - \delta < 0$$

Graphical Jacobian

Example



$$\begin{aligned}\frac{\partial f}{\partial x} &= - \\ \frac{\partial f}{\partial y} &= - \\ \frac{\partial g}{\partial x} &= + \\ \frac{\partial g}{\partial y} &= -\end{aligned}$$

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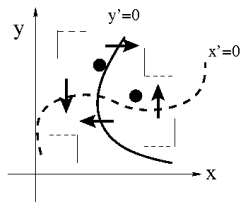
$$J = \begin{pmatrix} -\alpha & -\beta \\ +\gamma & -\delta \end{pmatrix} \quad \text{and}$$

$$\text{tr } J = -\alpha - \delta < 0$$

This is either a stable spiral or a stable node.

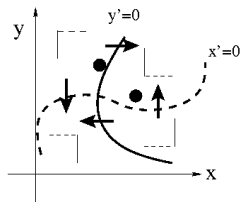
Graphical Jacobian

Example 2



Graphical Jacobian

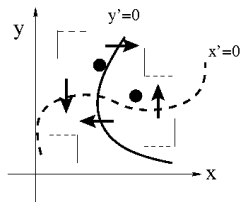
Example 2



$$\frac{\partial f}{\partial x} = +$$

Graphical Jacobian

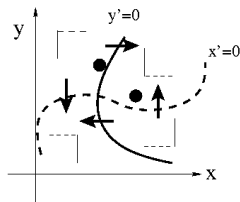
Example 2



$$\frac{\partial f}{\partial x} = +$$
$$\frac{\partial f}{\partial y} = +$$

Graphical Jacobian

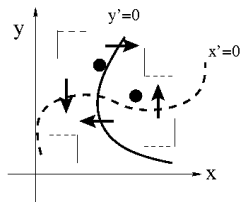
Example 2



$$\begin{aligned}\frac{\partial f}{\partial x} &= + \\ \frac{\partial f}{\partial y} &= + \\ \frac{\partial g}{\partial x} &= +\end{aligned}$$

Graphical Jacobian

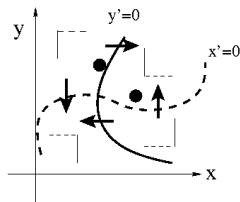
Example 2



$$\begin{aligned}\frac{\partial f}{\partial x} &= + \\ \frac{\partial f}{\partial y} &= + \\ \frac{\partial g}{\partial x} &= + \\ \frac{\partial g}{\partial y} &= -\end{aligned}$$

Graphical Jacobian

Example 2

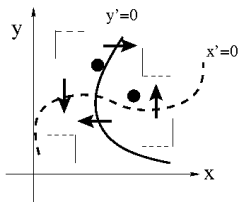


$$\begin{aligned}\frac{\partial f}{\partial x} &= + \\ \frac{\partial f}{\partial y} &= + \\ \frac{\partial g}{\partial x} &= + \\ \frac{\partial g}{\partial y} &= -\end{aligned}$$

$$J = \begin{pmatrix} +\alpha & +\beta \\ +\gamma & -\delta \end{pmatrix}$$

Graphical Jacobian

Example 2



$$\begin{aligned}\frac{\partial f}{\partial x} &= + \\ \frac{\partial f}{\partial y} &= + \\ \frac{\partial g}{\partial x} &= + \\ \frac{\partial g}{\partial y} &= -\end{aligned}$$

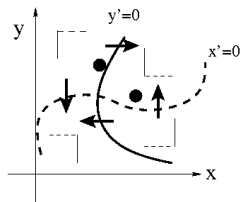
Therefore

$$J = \begin{pmatrix} +\alpha & +\beta \\ +\gamma & -\delta \end{pmatrix}$$

$$\det J = (+\alpha \times -\delta) - (\gamma \times \beta)$$

Graphical Jacobian

Example 2



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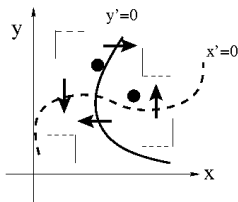
Therefore

$$J = \begin{pmatrix} +\alpha & +\beta \\ +\gamma & -\delta \end{pmatrix}$$

$$\det J = (+\alpha \times -\delta) - (\gamma \times \beta) < 0$$

Graphical Jacobian

Example 2



$$\begin{aligned}\frac{\partial f}{\partial x} &= + \\ \frac{\partial f}{\partial y} &= + \\ \frac{\partial g}{\partial x} &= + \\ \frac{\partial g}{\partial y} &= -\end{aligned}$$

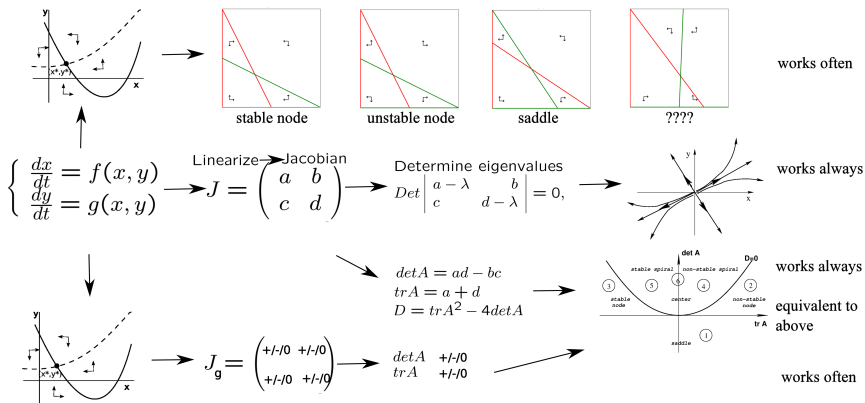
Therefore

$$J = \begin{pmatrix} +\alpha & +\beta \\ +\gamma & -\delta \end{pmatrix}$$

$$\det J = (+\alpha \times -\delta) - (\gamma \times \beta) < 0$$

So this equilibrium is a saddle point.

A Summary of Different Methods



Plan for Analysis

Now that we have learned these different methods:
when to use which method?

Plan for Analysis

Now that we have learned these different methods:
when to use which method? **From easy to more difficult!**

Plan for Analysis

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General method for determining equilibrium type:

Plan for Analysis

Now that we have learned these different methods:
when to use which method? **From easy to more difficult!**
General method for determining equilibrium type:

- 1 Draw null-clines and vectorfield, find equilibria.

Plan for Analysis

Now that we have learned these different methods:
when to use which method? **From easy to more difficult!**

General method for determining equilibrium type:

- 1 Draw null-clines and vectorfield, find equilibria.
- 2 Look at vectorfield to determine equilibrium type.

Plan for Analysis

Now that we have learned these different methods:
when to use which method? **From easy to more difficult!**

General method for determining equilibrium type:

- 1 Draw null-clines and vectorfield, find equilibria.
- 2 Look at vectorfield to determine equilibrium type.
- 3 If 2 fails: Use graphical jacobian to determine equilibrium type (using the determinant-trace method).

Plan for Analysis

Now that we have learned these different methods:
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General method for determining equilibrium type:

- 1 Draw null-clines and vectorfield, find equilibria.
- 2 Look at vectorfield to determine equilibrium type.
- 3 If 2 fails: Use graphical jacobian to determine equilibrium type (using the determinant-trace method).
- 4 If 3 fails: Use analytical jacobian to determine equilibrium type (using determinant-trace or eigenvalue method).

Plan for Analysis

Now that we have learned these different methods:
when to use which method? **From easy to more difficult!**

General method for determining equilibrium type:

- 1 Draw null-clines and vectorfield, find equilibria.
- 2 Look at vectorfield to determine equilibrium type.
- 3 If 2 fails: Use graphical jacobian to determine equilibrium type (using the determinant-trace method).
- 4 If 3 fails: Use analytical jacobian to determine equilibrium type (using determinant-trace or eigenvalue method).

Finally, connect local phase portrait of equilibria to get **global phase portrait** and determine basins of attraction.

Example

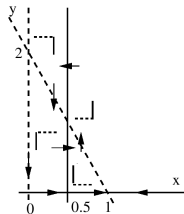
Let us use this approach for:

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

(We have seen this system in previous chapters)

Example

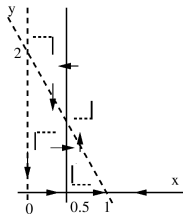
Null-clines and vectorfield:



We see three equilibria: $(0, 0)$, $(0.5, 1)$ and $(1, 0)$

Example

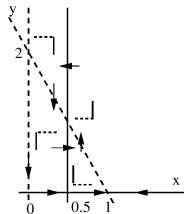
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 $(0, 0)$: one converging one diverging direction: saddle

Example

Null-clines and vectorfield:



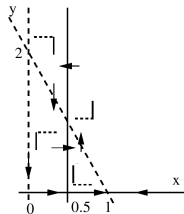
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Example

Null-clines and vectorfield:



We see three equilibria: $(0, 0)$, $(0.5, 1)$ and $(1, 0)$

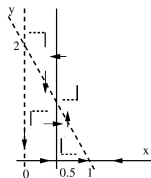
$(0, 0)$: one converging one diverging direction: saddle

$(1, 0)$: one converging one diverging direction: saddle

$(0.5, 1)$: rotation, equilibrium type unclear

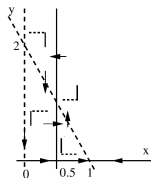
Example

Graphical Jacobian:



Example

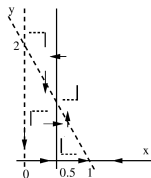
Graphical Jacobian:



$$(0,0): J = \begin{pmatrix} +\alpha & 0 \\ 0 & -\delta \end{pmatrix}$$

Example

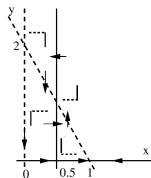
Graphical Jacobian:



$$(0,0): J = \begin{pmatrix} +\alpha & 0 \\ 0 & -\delta \end{pmatrix}, \det J = -\alpha\delta < 0:$$

Example

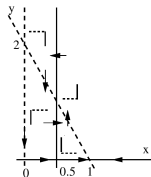
Graphical Jacobian:



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Example

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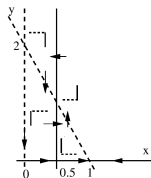


$$(0,0): J = \begin{pmatrix} +\alpha & 0 \\ 0 & -\delta \end{pmatrix}, \det J = -\alpha\delta < 0: \text{ saddle}$$

$$(1,0): J = \begin{pmatrix} -\alpha & -\beta \\ 0 & +\delta \end{pmatrix}$$

Example

Graphical Jacobian:

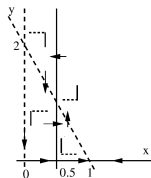


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Example

Graphical Jacobian:

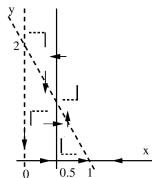


$$(0, 0): J = \begin{pmatrix} +\alpha & 0 \\ 0 & -\delta \end{pmatrix}, \det J = -\alpha\delta < 0: \text{ saddle}$$

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Example

Graphical Jacobian:



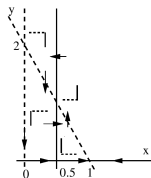
$$(0,0): J = \begin{pmatrix} +\alpha & 0 \\ 0 & -\delta \end{pmatrix}, \det J = -\alpha\delta < 0: \text{ saddle}$$

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$$(0.5,1): J = \begin{pmatrix} -\alpha & -\beta \\ +\gamma & 0 \end{pmatrix}$$

Example

Graphical Jacobian:



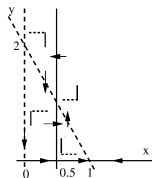
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$$(0.5, 1): J = \begin{pmatrix} -\alpha & -\beta \\ +\gamma & 0 \end{pmatrix}, \det J = \gamma\beta > 0$$

Example

Graphical Jacobian:



$$(0,0): J = \begin{pmatrix} +\alpha & 0 \\ 0 & -\delta \end{pmatrix}, \det J = -\alpha\delta < 0: \text{ saddle}$$

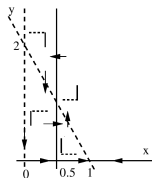
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$$\text{tr } J = -\alpha < 0:$$

Example

Graphical Jacobian:



$$(0,0): J = \begin{pmatrix} +\alpha & 0 \\ 0 & -\delta \end{pmatrix}, \det J = -\alpha\delta < 0: \text{ saddle}$$

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$$(0.5,1): J = \begin{pmatrix} -\alpha & -\beta \\ +\gamma & 0 \end{pmatrix}, \det J = \gamma\beta > 0$$

$\text{tr } J = -\alpha < 0$: stable spiral or node

Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Analytical Jacobian:

$(0, 0)$:

$(1, 0)$:

$(0.5, 1)$:

Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Analytical Jacobian:

$$(0,0): J_1 = \begin{pmatrix} 3 & 0 \\ 0 & -0.25 \end{pmatrix}$$

(1,0):

(0.5, 1):

Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Analytical Jacobian:

$$(0,0): J_1 = \begin{pmatrix} 3 & 0 \\ 0 & -0.25 \end{pmatrix}, \det J_1 = 3 \times -0.25 < 0$$

(1,0):

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Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Analytical Jacobian:

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(1,0):

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Example

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$$(1,0): J_2 = \begin{pmatrix} -3 & -1.5 \\ 0 & 0.25 \end{pmatrix}$$

$$(0.5, 1):$$

Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

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Example

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Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

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$$(1,0): J_2 = \begin{pmatrix} -3 & -1.5 \\ 0 & 0.25 \end{pmatrix}, \det J_2 = -3 \times 0.25 < 0: \text{ saddle}$$

$$(0.5,1): J_3 = \begin{pmatrix} -1.5 & -0.75 \\ 0.5 & 0 \end{pmatrix}$$

Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Analytical Jacobian:

$$(0,0): J_1 = \begin{pmatrix} 3 & 0 \\ 0 & -0.25 \end{pmatrix}, \det J_1 = 3 \times -0.25 < 0: \text{ saddle}$$

$$(1,0): J_2 = \begin{pmatrix} -3 & -1.5 \\ 0 & 0.25 \end{pmatrix}, \det J_2 = -3 \times 0.25 < 0: \text{ saddle}$$

$$(0.5,1): J_3 = \begin{pmatrix} -1.5 & -0.75 \\ 0.5 & 0 \end{pmatrix}, \\ \det J_3 = -0.5 \times -0.75 = 0.375 > 0$$

Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Analytical Jacobian:

$$(0,0): J_1 = \begin{pmatrix} 3 & 0 \\ 0 & -0.25 \end{pmatrix}, \det J_1 = 3 \times -0.25 < 0: \text{ saddle}$$

$$(1,0): J_2 = \begin{pmatrix} -3 & -1.5 \\ 0 & 0.25 \end{pmatrix}, \det J_2 = -3 \times 0.25 < 0: \text{ saddle}$$

$$(0.5,1): J_3 = \begin{pmatrix} -1.5 & -0.75 \\ 0.5 & 0 \end{pmatrix},$$
$$\det J_3 = -0.5 \times -0.75 = 0.375 > 0,$$
$$\text{tr } J_3 = -1.5 < 0 \text{ and}$$

Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Analytical Jacobian:

$$(0,0): J_1 = \begin{pmatrix} 3 & 0 \\ 0 & -0.25 \end{pmatrix}, \det J_1 = 3 \times -0.25 < 0: \text{ saddle}$$

$$(1,0): J_2 = \begin{pmatrix} -3 & -1.5 \\ 0 & 0.25 \end{pmatrix}, \det J_2 = -3 \times 0.25 < 0: \text{ saddle}$$

$$(0.5,1): J_3 = \begin{pmatrix} -1.5 & -0.75 \\ 0.5 & 0 \end{pmatrix},$$
$$\det J_3 = -0.5 \times -0.75 = 0.375 > 0,$$
$$\text{tr } J_3 = -1.5 < 0 \text{ and}$$
$$D = (-1.5)^2 - 4 \times 0.375 = 0.75 > 0$$

Example

$$\begin{cases} \frac{dx}{dt} = 3x(1-x) - 1.5xy \\ \frac{dy}{dt} = 0.5xy - 0.25y \end{cases}$$

Analytical Jacobian:

$$(0,0): J_1 = \begin{pmatrix} 3 & 0 \\ 0 & -0.25 \end{pmatrix}, \det J_1 = 3 \times -0.25 < 0: \text{ saddle}$$

$$(1,0): J_2 = \begin{pmatrix} -3 & -1.5 \\ 0 & 0.25 \end{pmatrix}, \det J_2 = -3 \times 0.25 < 0: \text{ saddle}$$

$$(0.5,1): J_3 = \begin{pmatrix} -1.5 & -0.75 \\ 0.5 & 0 \end{pmatrix},$$

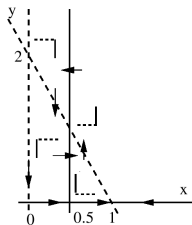
$$\det J_3 = -0.5 \times -0.75 = 0.375 > 0,$$

$$\text{tr } J_3 = -1.5 < 0 \text{ and}$$

$$D = (-1.5)^2 - 4 \times 0.375 = 0.75 > 0: \text{ stable node}$$

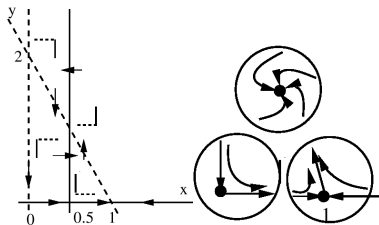
Example

Local phase portraits around equilibria and global one:



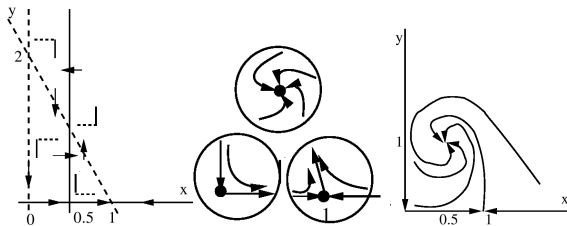
Example

Local phase portraits around equilibria and global one:



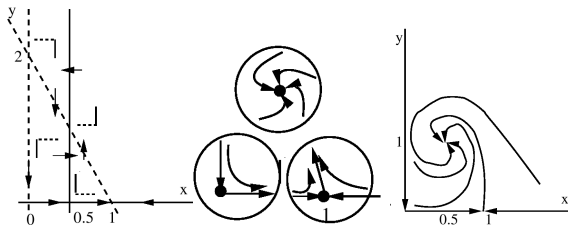
Example

Local phase portraits around equilibria and global one:



Example

Local phase portraits around equilibria and global one:



Single large basin of attraction for $(0.5, 1)$.

A backup plan

We know that linearization, Jacobians, etc. are hard!

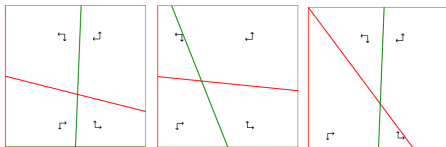
No need to panic: null-clines and vectorfield often enough.

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We can even push the use of vectorfields a bit further:

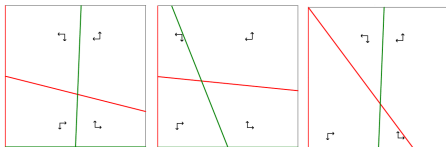


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No need to panic: null-clines and vectorfield often enough.

We can even push the use of vectorfields a bit further:



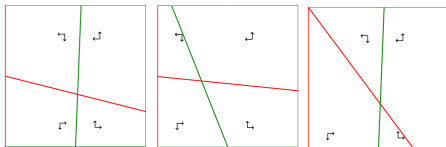
- add x : does x grow or decline?

A backup plan

We know that linearization, Jacobians, etc. are hard!

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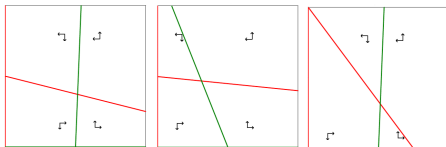
- add x : does x grow or decline?
 - growth: unstable, decline: stable

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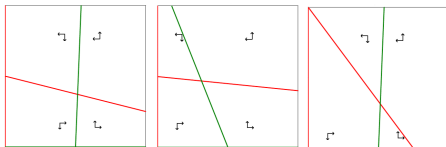
- add x : does x grow or decline?
 - growth: unstable, decline: stable
- add y : does y grow or decline?
 - growth: unstable, decline: stable

A backup plan

We know that linearization, Jacobians, etc. are hard!

No need to panic: null-clines and vectorfield often enough.

We can even push the use of vectorfields a bit further:



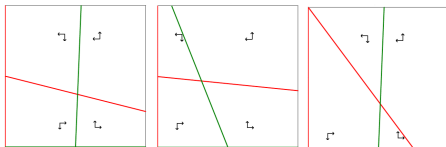
- add x : does x grow or decline?
 - growth: unstable, decline: stable
- add y : does y grow or decline?
 - growth: unstable, decline: stable
- If 2 stable directions: stable!

A backup plan

We know that linearization, Jacobians, etc. are hard!

No need to panic: null-clines and vectorfield often enough.

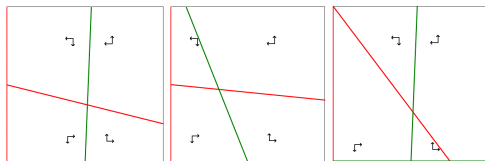
We can even push the use of vectorfields a bit further:



- add x : does x grow or decline?
 - growth: unstable, decline: stable
- add y : does y grow or decline?
 - growth: unstable, decline: stable
- If 2 stable directions: stable!
- If 1 or 2 unstable directions: unstable!

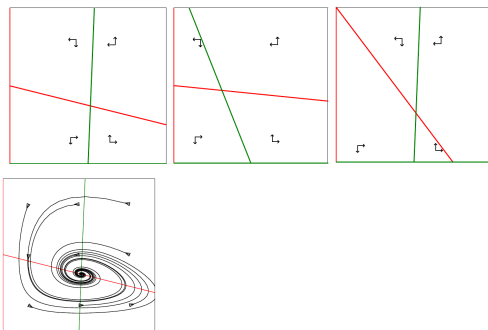
A backup plan

Note that this approach is close to the graphical Jacobian approach!



A backup plan

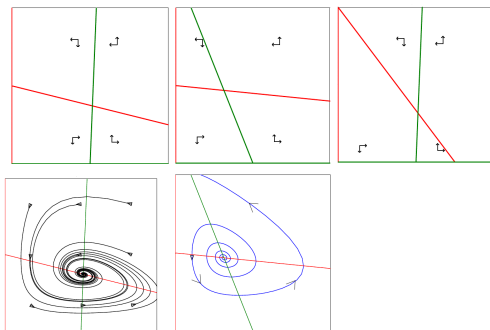
Note that this approach is close to the graphical Jacobian approach!



Case 1: add x : decline, add y decline: stable!

A backup plan

Note that this approach is close to the graphical Jacobian approach!

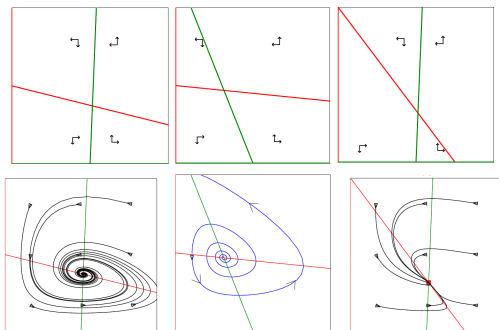


Case 1: add x : decline, add y decline: stable!

Case 2: add x : decline, add y increase: unstable!

A backup plan

Note that this approach is close to the graphical Jacobian approach!



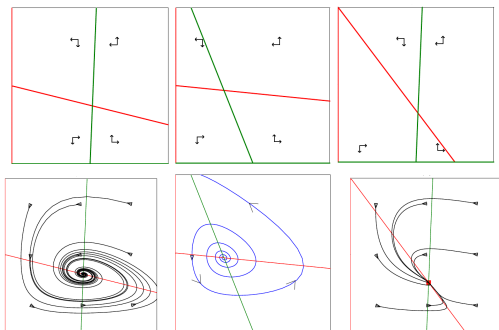
Case 1: add x : decline, add y decline: stable!

Case 2: add x : decline, add y increase: unstable!

Case 3: add x : decline, add y decline: stable!

A backup plan

Note that this approach is close to the graphical Jacobian approach!



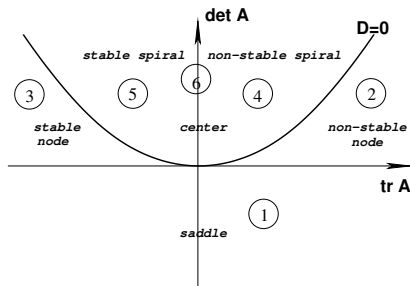
Case 1: add x : decline, add y decline: stable!

Case 2: add x : decline, add y increase: unstable!

Case 3: add x : decline, add y decline: stable!

But....does not work for horizontal/vertical nullclines!

Plan of Analysis



- 1 Draw null-clines and vectorfield, find equilibria.
- 2 Look at vectorfield to determine equilibrium type.
- 3 If 2 fails: Use graphical jacobian to determine equilibrium type (using the determinant-trace method).
- 4 If 3 fails: Use analytical jacobian to determine equilibrium type (using determinant-trace or eigenvalue method).